

Decoding the Card Trick: Autonomous Strategy Discovery via Reinforcement Learning

Meenakshisundram Ganapathi Subramanian
Master's Thesis

Chair: Dr. Zhe Xu

Members: Dr. Spring Berman
Dr. Hamidreza Marvi

Introduction: The Challenge of Uncertainty



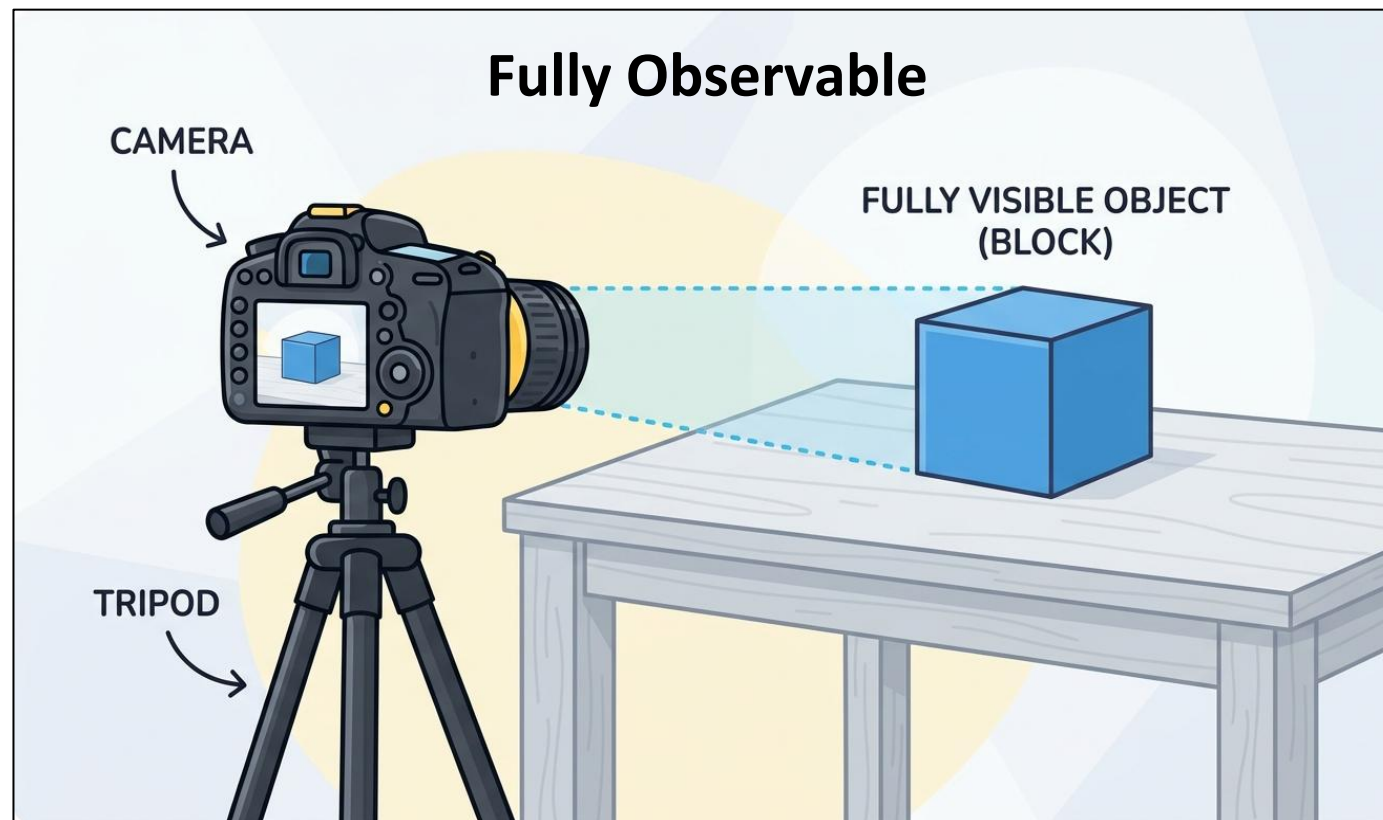
The Shift: Structured industry



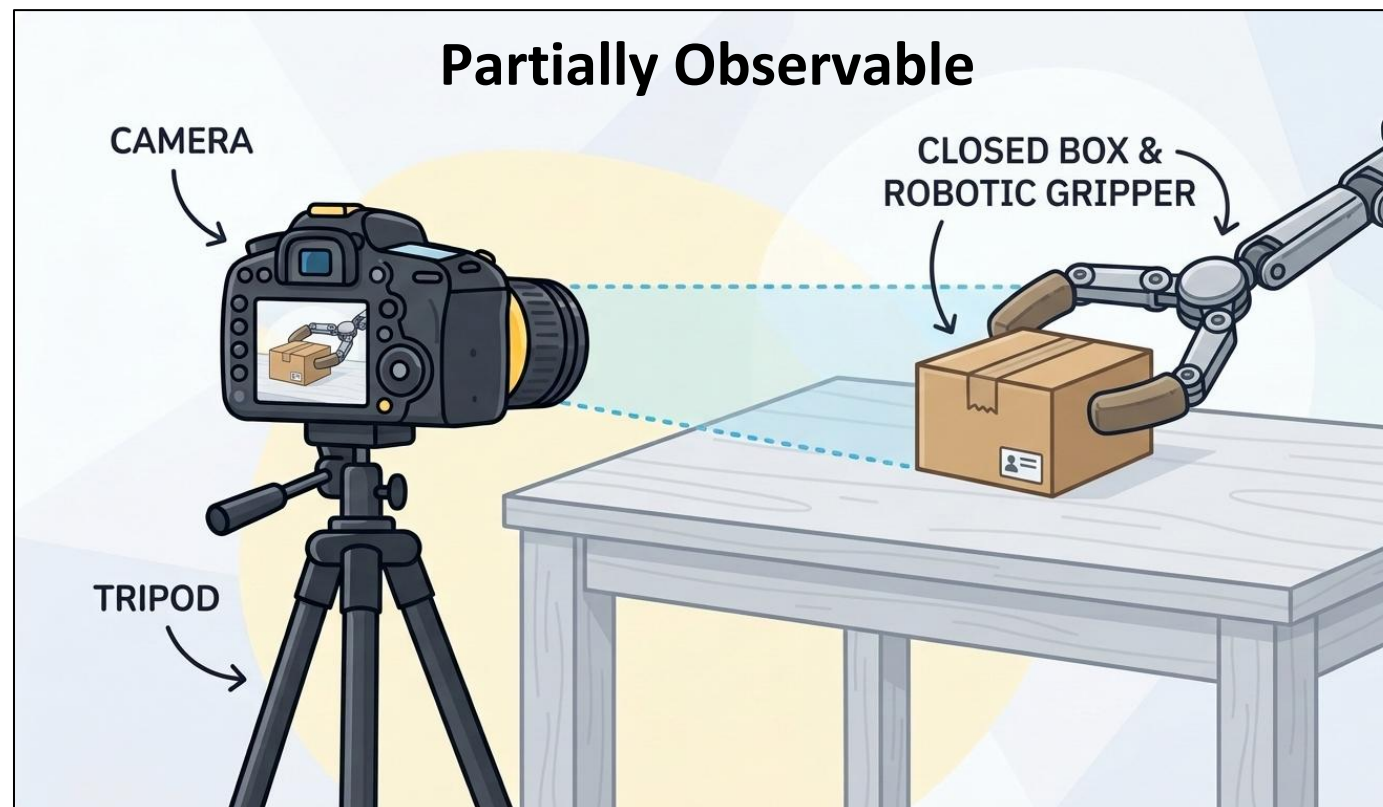
The Obstacle: Hidden variables

The Goal: Autonomous Information Gathering

The Technical Bottleneck: Partial Observability



- **The Limit of Passive Vision:**
Cameras alone cannot solve physical occlusion.
- **The Need for Active Perception:**
Hidden states demand exploratory action.



- **Action Precedes Computation:**
The system must physically manipulate the environment to reveal the missing data.

The Benchmark: Three-Card Deception



- **Beyond a Parlor Trick:**
This deception serves as a mathematically rigorous proxy for unstructured real-world tasks.
- **The Rules of the Environment:**
A known target card is subjected to a deterministic, hidden spatial permutation.
- **The Objective:**
The robotic system must logically deduce the target's final hidden location with 100% autonomy, without relying on any human-coded heuristics.

Mechanics Step 1: The Initial State

Step 1: The Eyes (Initial State)



Observer registers starting positions: [A♠, K♥, Q♦].

- **Establishing Ground Truth:**
Before any action occurs, the system must register the board, which is done via terminal initialization.
- **The State Vector:**
Each card is mapped to a specific spatial index.
- **Example State:**
[Index 0: A♠,
Index 1: K♥,
Index 2: Q♦]

Mechanics Step 2: The Action

Step 2: The Action (Hidden Swap)

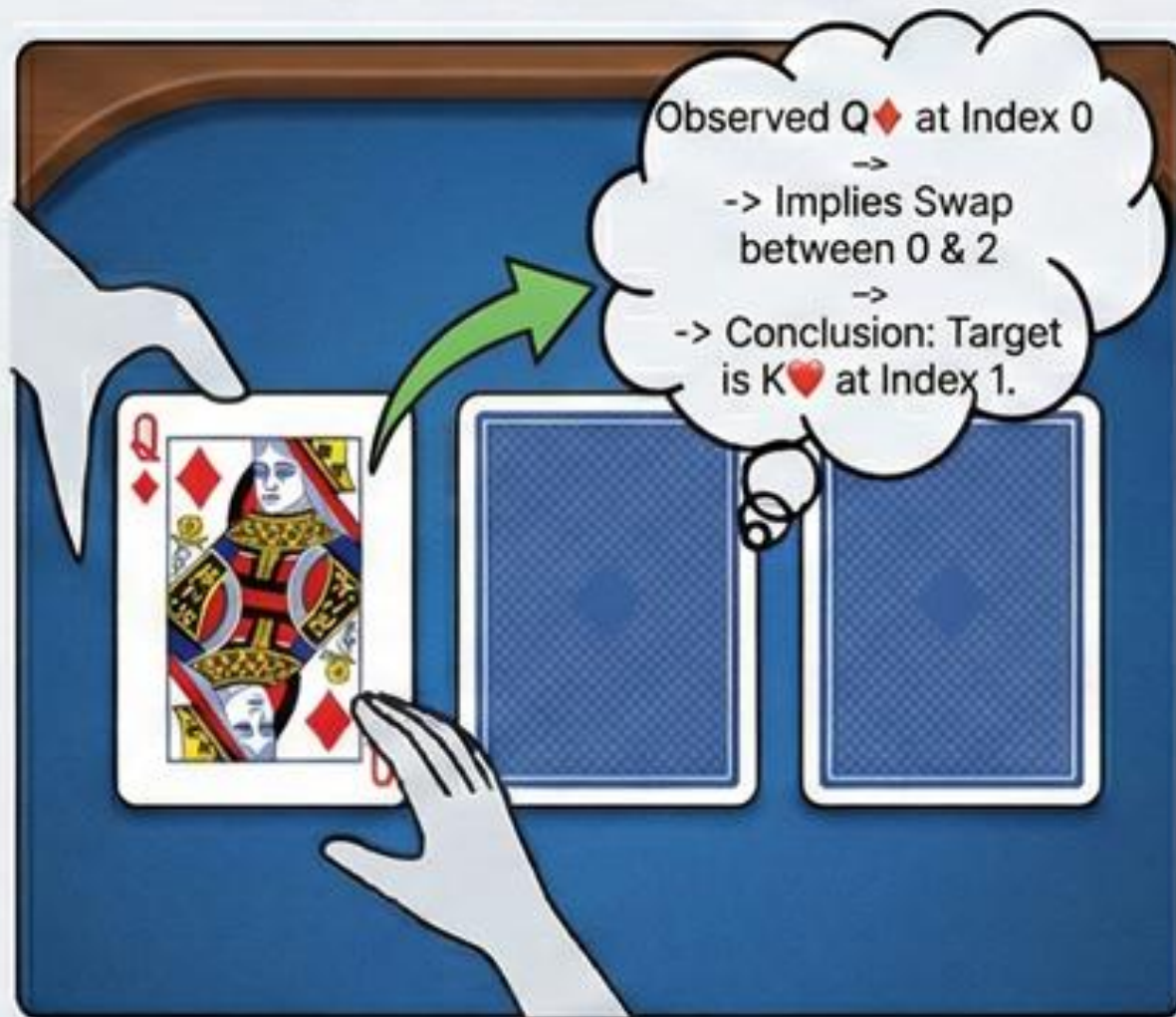


User picks K♥ (Target),
swaps the others (A♠ & Q♦).

- **The Blind State:**
Cards are face-down.
- **Deterministic Swap:**
The user's target stays; the other two cards switch positions.
- **The Result:**
The initial state is invalidated. The target is now a *hidden variable*.

Mechanics Step 3: The Deduction

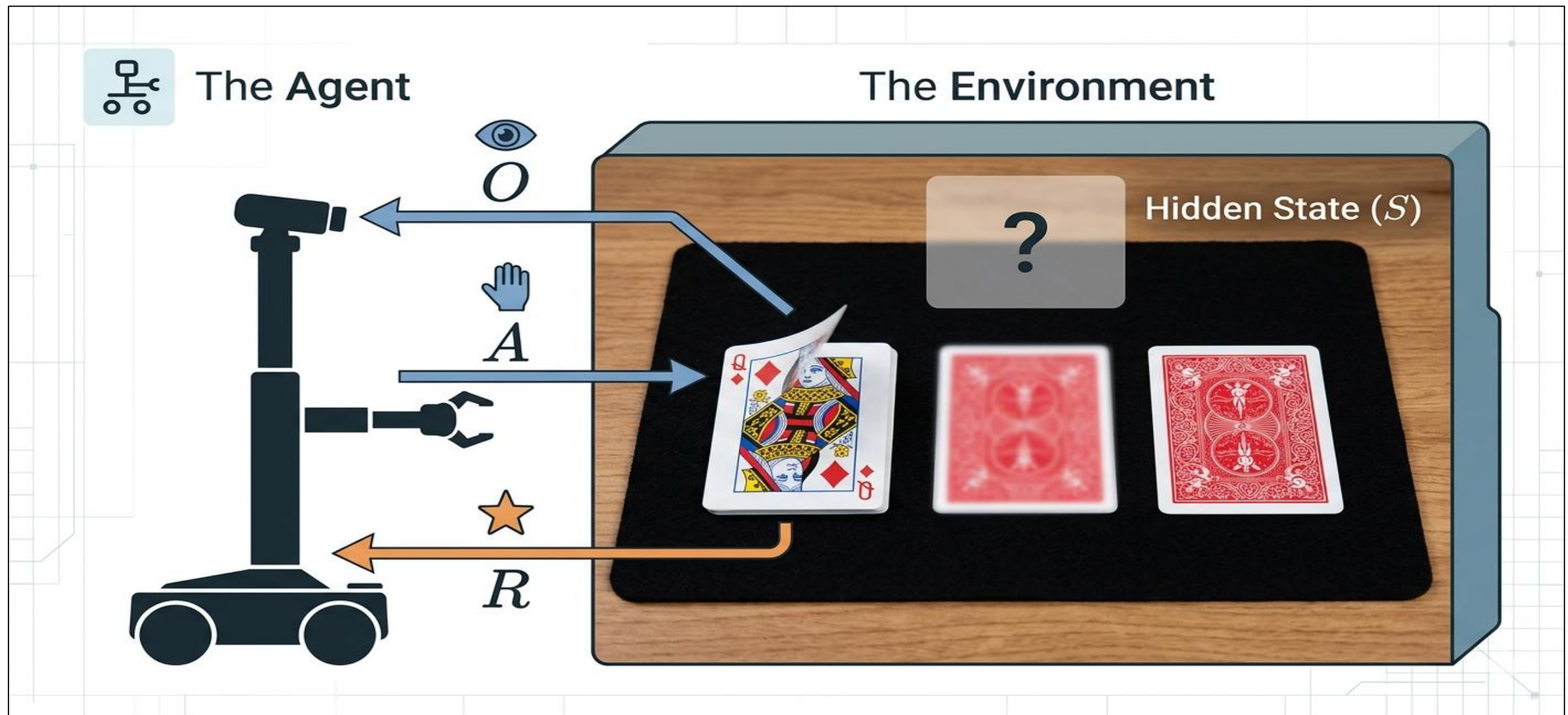
Step 3: The Brain (Deduction)



Observer flips Index 0, sees Q♦,
deduces K♥ is the un-moved
Target.

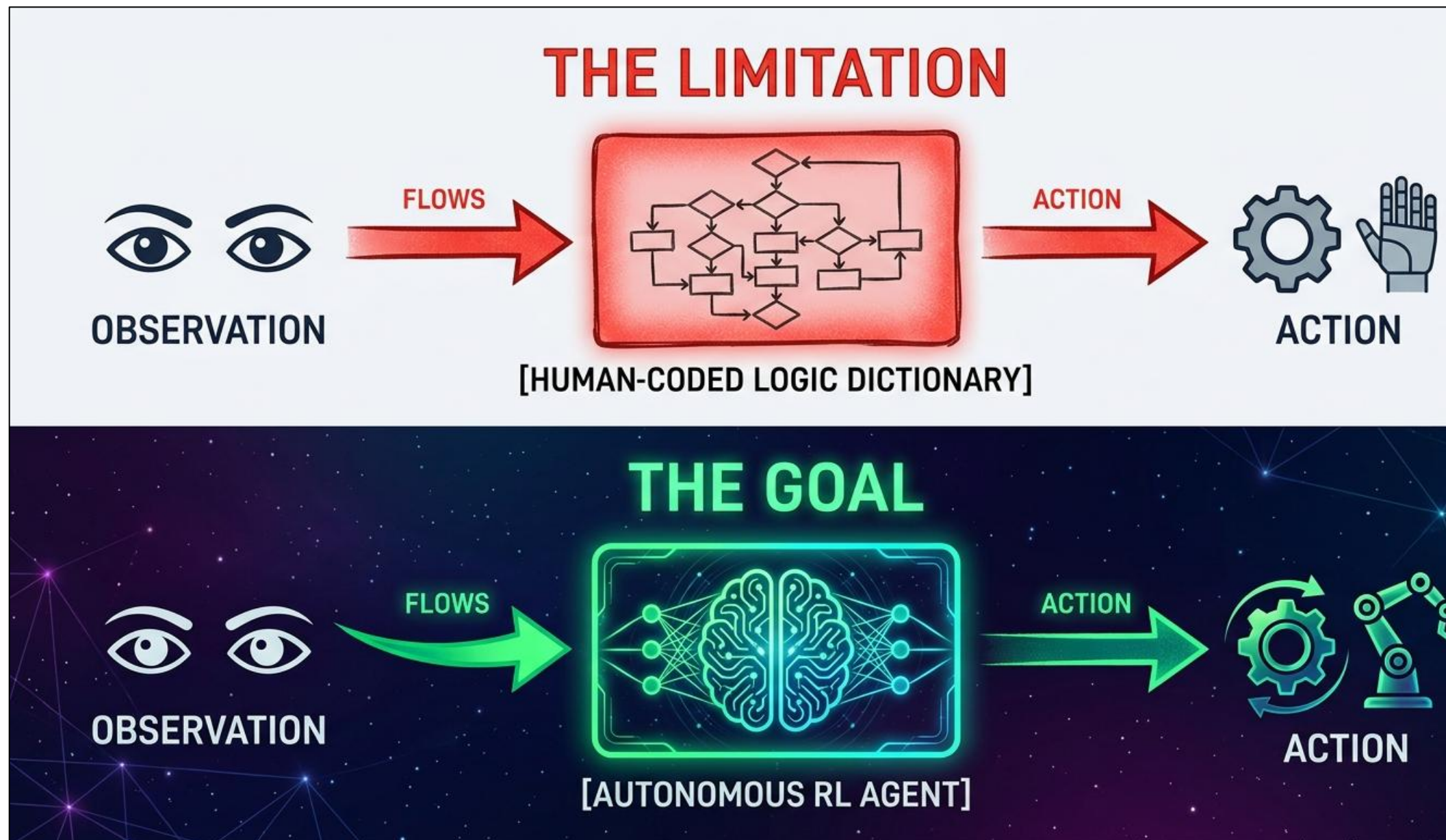
- **Physical Action:**
The robot flips one card to break the ambiguity.
- **Visual Observation:**
The camera registers the revealed card (e.g., seeing Q♦).
- **Logical Deduction:**
The system compares this observation to the initial state to definitively locate the hidden target.

Problem Formulation: The POMDP



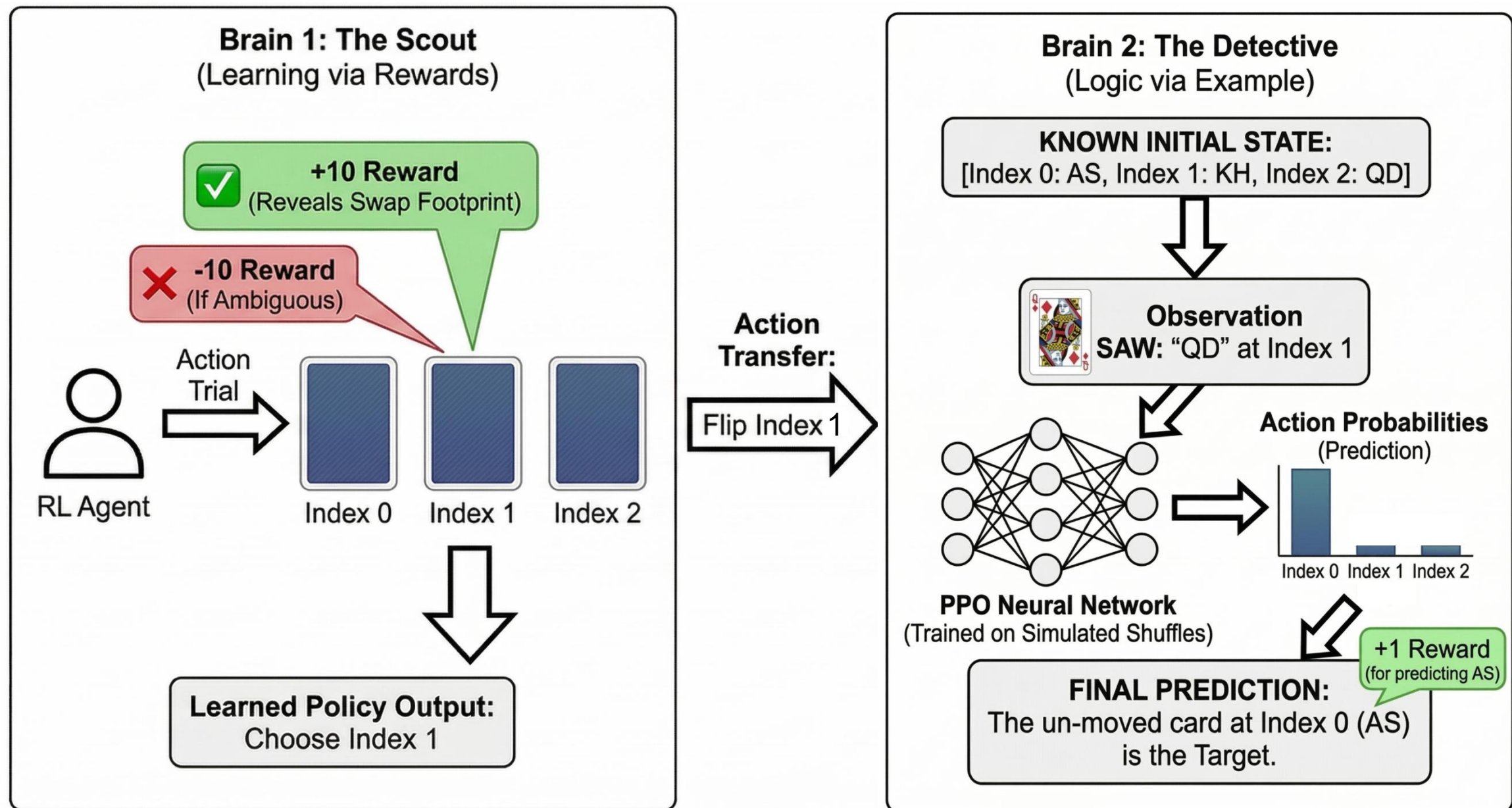
- **State Space (S):**
The agent relies on a continuous dummy variable ($s = [0.0]$).
- **Action Space (A):**
Discrete physical manipulation (index choice).
- **Observation (O):**
The vision pipeline registering the newly revealed card.
- **Reward Function (R):**
Evaluated via the "Multiverse" simulation.

State of the Art & The Gap



- **Reinforcement Learning:** High-level task planning.
- **Active Perception:** Resolving POMDPs.
- **YOLO & LLMs:** Robust vision and dynamic interaction.

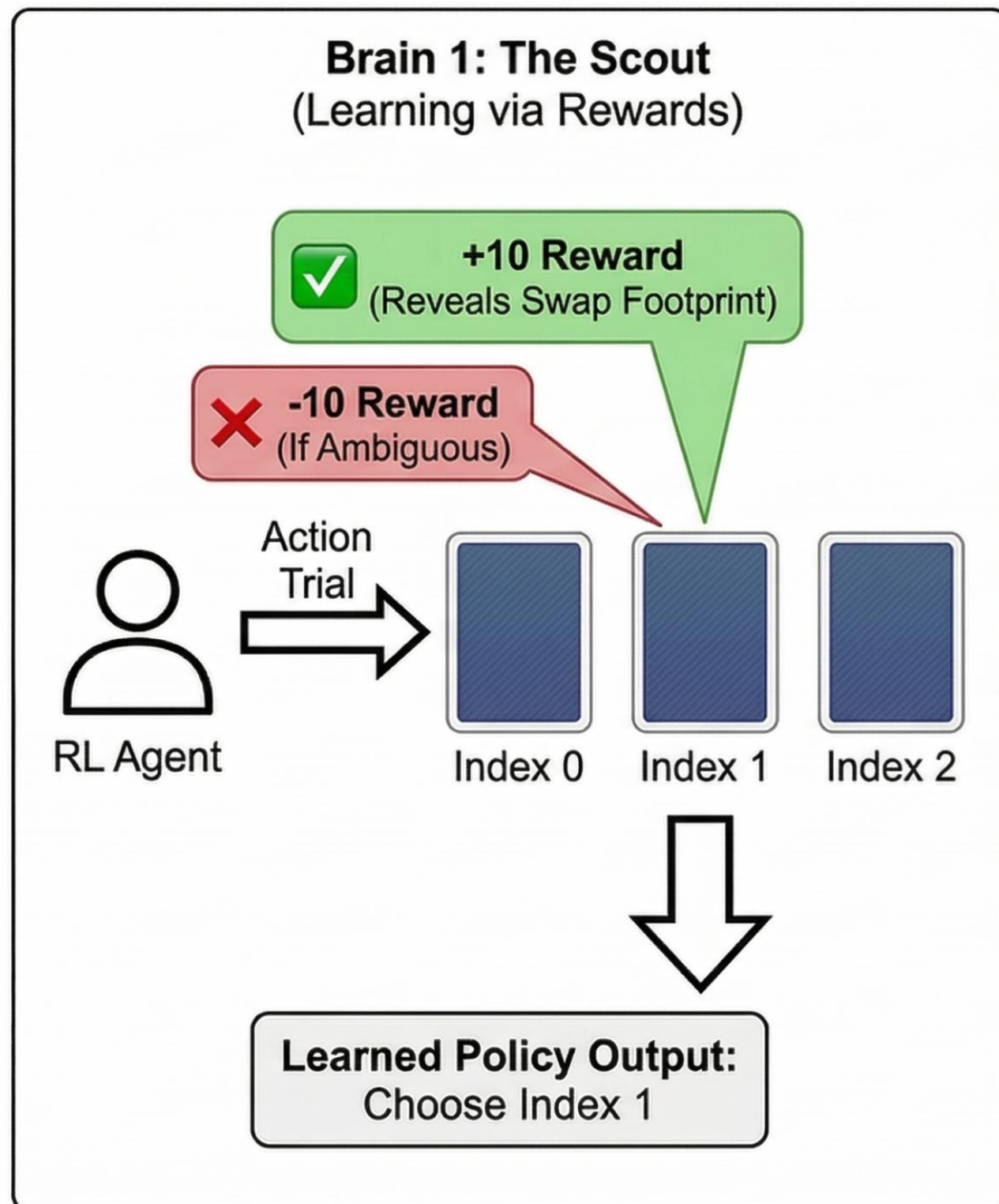
Methodology: Decoupled RL Architecture



Splitting the cognitive load

- Stage 1: Strategy Discovery.
- Stage 2: State Inference.

Stage 1: Strategy Discovery (The Scout)



- **Environment Setup:**

A custom Gymnasium environment that mimics the deterministic swap mechanics.

- **Action Space (A):**

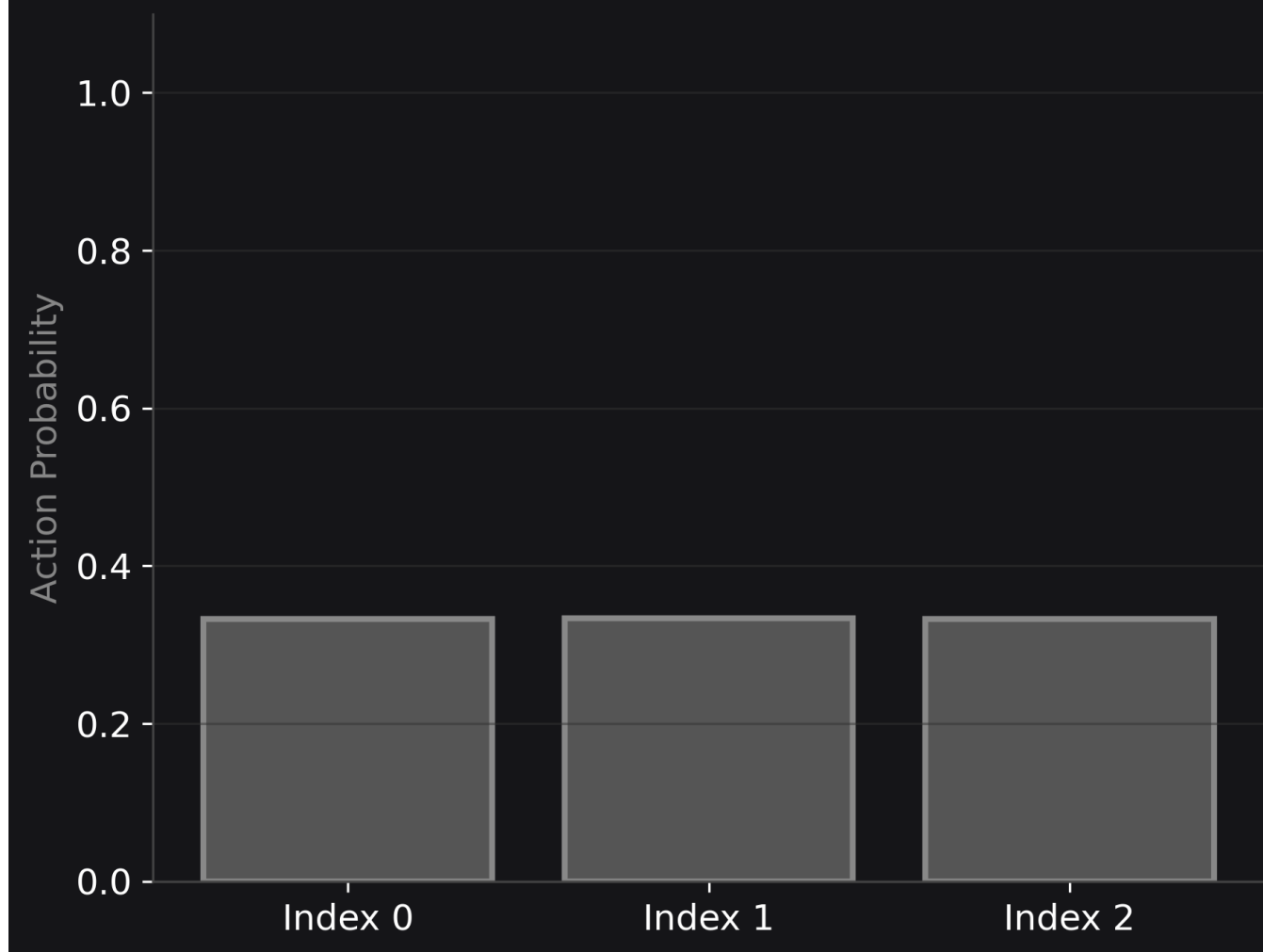
Discrete choice of which card index to physically manipulate.

- **The Discovery Goal:**

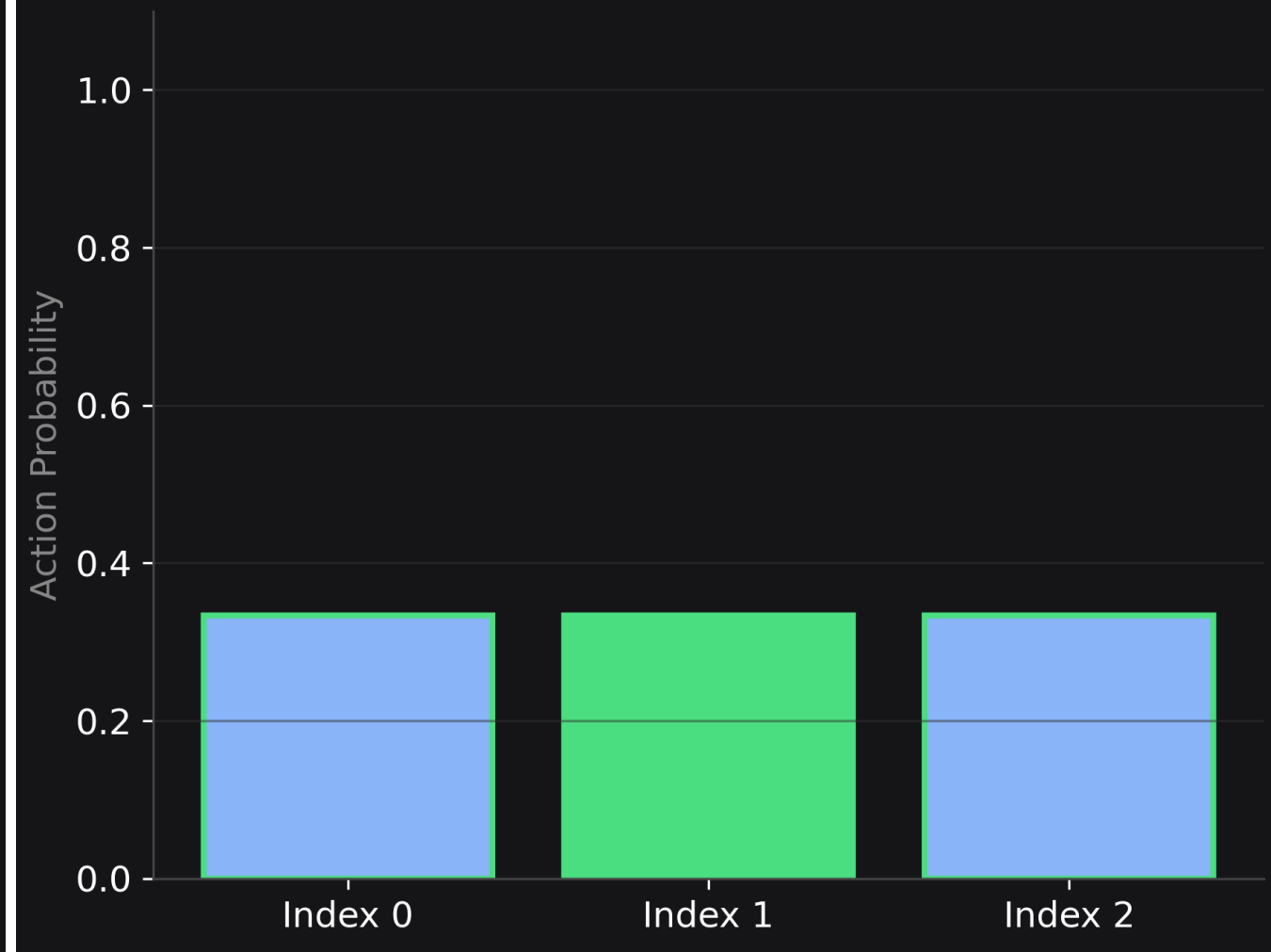
Learning to act optimally through pure environmental interaction, without pre-defined heuristics.

Training the Scout: Policy Convergence

Timestep 0: Random Exploration

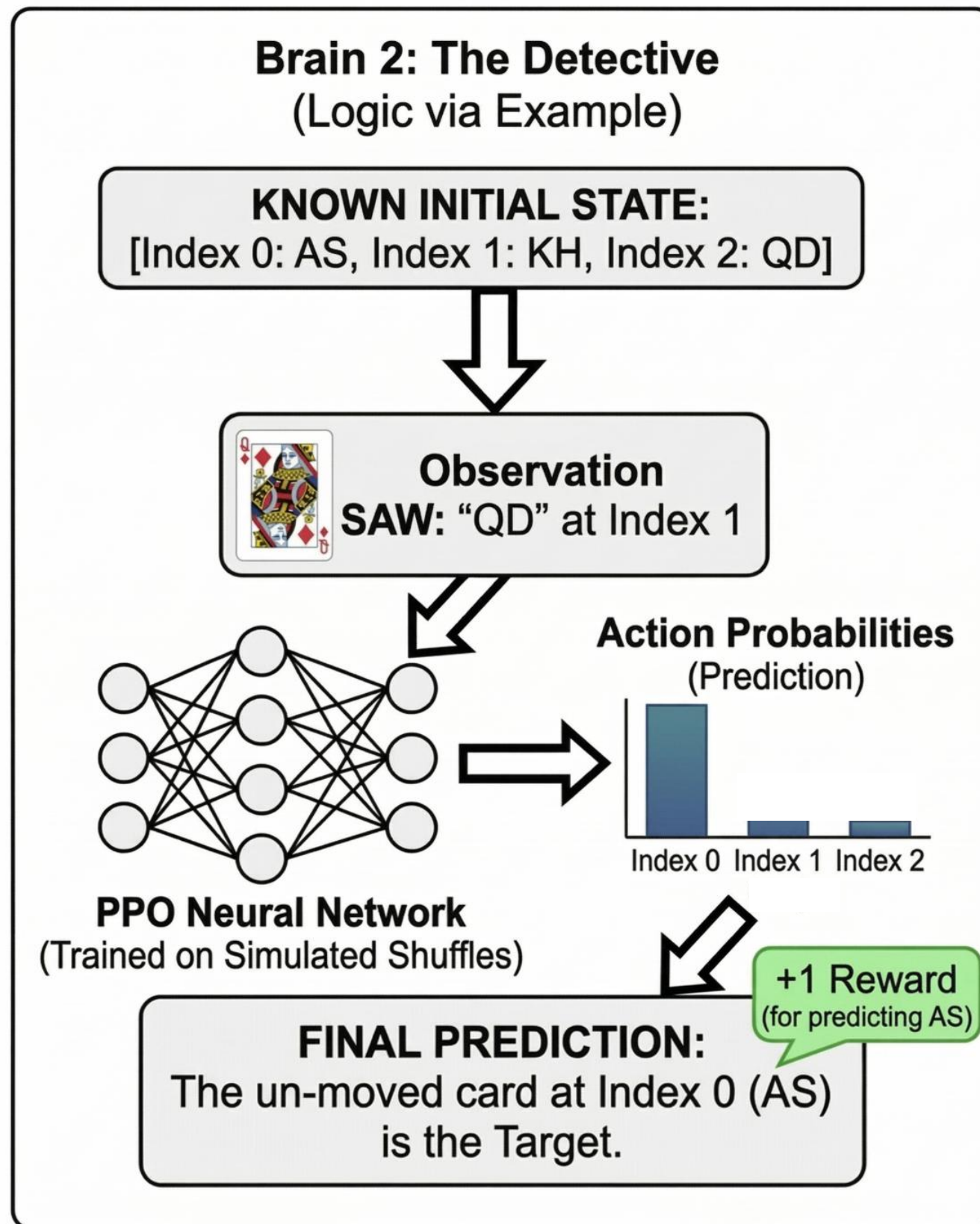


Timestep 5000: Strategy Converged



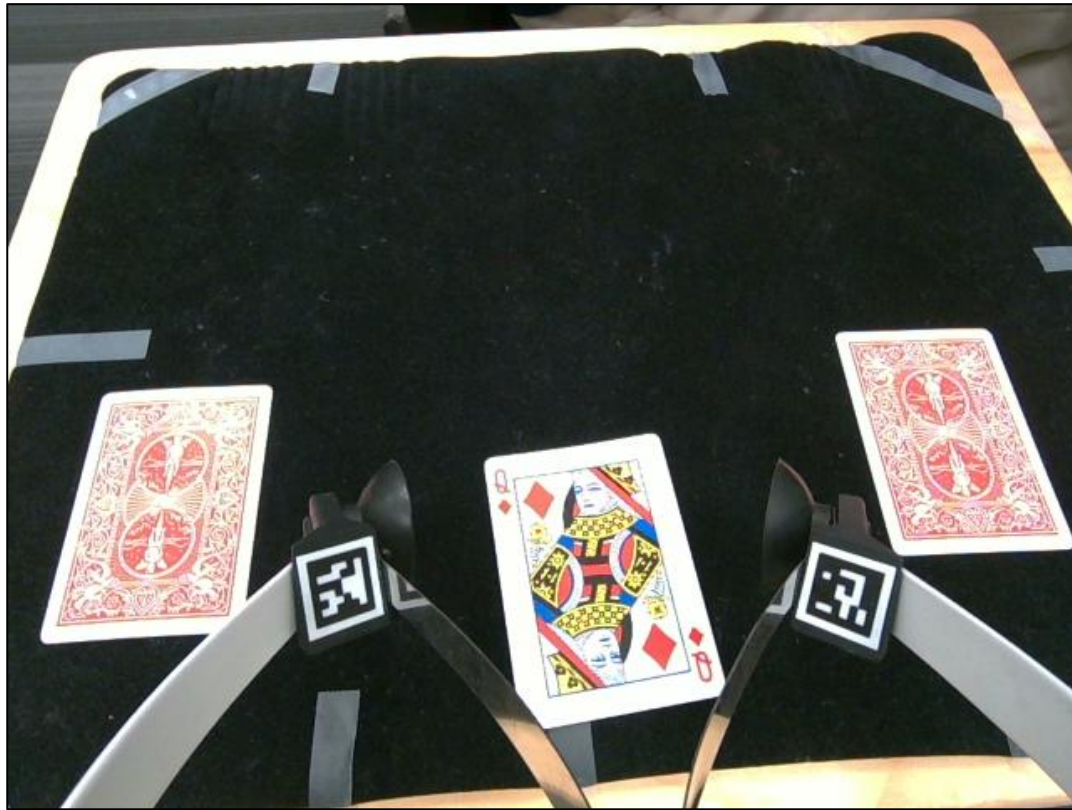
- PPO Agent evaluates the "Multiverse" reward structure.
- Policy converges in <5,000 timesteps.
- Autonomously optimizes for physical kinematics (Index 1).

Stage 2: State Inference (The Detective)



- **The Goal:**
Reverse the swap mapping
(Observations → Hidden State).
- **Input State:**
Card identity at the optimal
sensing index.
- **Action Space:**
Discrete prediction of the user's
target.
- **Reward:**
Sparse feedback (+1.0 for
correct, -1.0 for incorrect)

The Perception Layer: Bridging Vision and Logic



```
hello-robot@stretch-re2-2013:~/ros2_ws$ ./run_detector.sh
Setting up ROS 2 environment...
Activating Python virtual environment...
Starting detector node...
[INFO] [1775096406.274208351] [card_detector_node]: Card Detector Node starting...
[INFO] [1775096406.275787454] [card_detector_node]: CODE_PATH=/home/hello-robot/ros2_ws/install/card_detector/lib/python3.10/site-packages/card_detector/detector_node.py | log_on_change_only=True | min_stable_frames=3
/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/node.py:440: UserWarning: when declaring parameter named 'tts_phrase_template', declaring a parameter only providing its name is deprecated. You have to either:
  - Pass a name and a default value different to "PARAMETER NOT SET" (and optionally a descriptor).
  - Pass a name and a parameter type.
  - Pass a name and a descriptor with `dynamic_typing=True`
  warnings.warn(
[INFO] [1775096406.281538280] [card_detector_node]: Loading YOLO model from: /home/hello-robot/ros2_ws/install/card_detector/share/card_detector/models/best.pt
[INFO] [1775096406.363018797] [card_detector_node]: Subscribed to: /camera/camera/color/image_rect_raw | conf>=0.60 | device=auto | debug=True | tts=on (qTTS only)
[INFO] [1775096407.289969433] [card_detector_node]: Detected: QD (conf=0.95)
```

- **Hardware:**

Intel RealSense D405 mounted on the Stretch 3 gripper for localized observation.

- **Perception:**

YOLO model fine-tuned on a hybrid dataset.

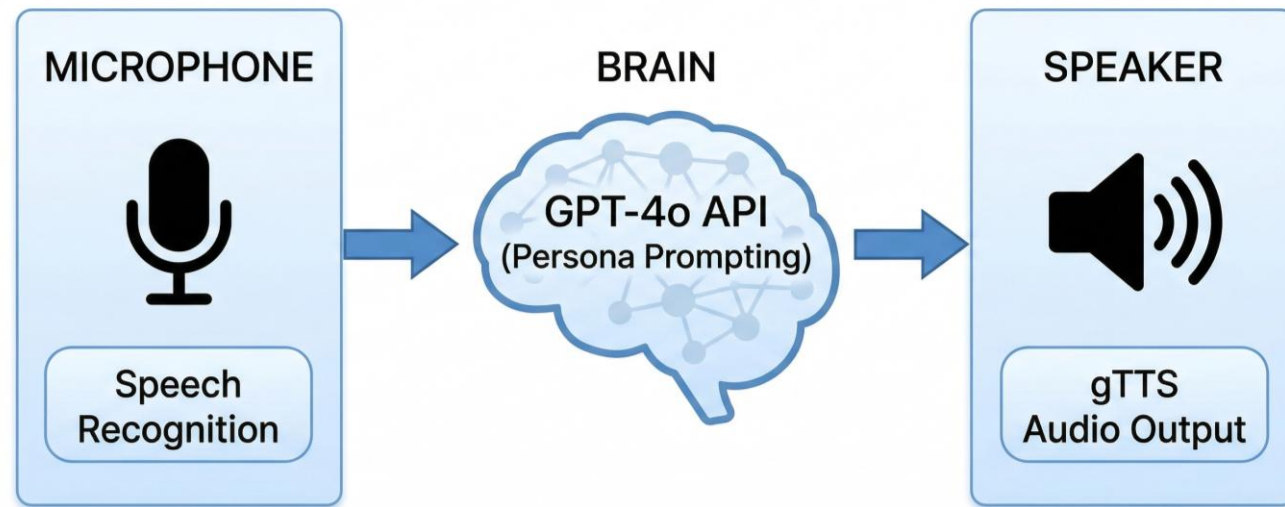
- **Software Architecture:**

ROS 2 deployed for asynchronous, real-time perception.

- **Performance Metric:**

High-confidence target lock (e.g., 95% on the Queen of Diamonds).

Physical Execution & Human Robot Interaction



Dynamic Dialogue:

- Asynchronous GPT-4o integration for persona-driven interaction.
- Non-blocking gTTS audio pipeline.

Dexterous Manipulation:

- Pre-calibrated physical trajectories via the `stretch_body` API.
- Precise sequential control of lift, wrist pitch/roll, and gripper.

System Architecture & Hardware Integration

```
hello-robot@stretch-re2-2013:~/ros2_ws$ /run_brain.sh 'AS' 'KH' 'QD'
Setting up ROS 2 environment...
Activating Python virtual environment...
Starting Brain (Logic Node) with cards:
LEFT: AS
MIDDLE: KH
RIGHT: QD
/home/hello-robot/.local/lib/python3.10/site-packages/pandas/core/computation/expressions.py:21:
UserWarning: Pandas requires version '2.8.4' or newer of 'numexpr' (version '2.8.1' currently installed).
  from pandas.core.computation.check import NUMEXPR_INSTALLED
/home/hello-robot/.local/lib/python3.10/site-packages/pandas/core/arrays/masked.py:61: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.2' currently installed).
  .
  from pandas.core import (
Disabling PyTorch because PyTorch >= 2.4 is required but found 2.1.0+cu118
PyTorch was not found. Models won't be available and only tokenizers, configuration and file/data
utilities can be used.
[INFO] [1775096461.937739245] [trick_logic_node_rl]: Brain 1 (Scout) Loaded.
[INFO] [1775096462.051127001] [trick_logic_node_rl]: Brain 2 (Detective) Loaded.
[INFO] [1775096565.013635157] [trick_logic_node_rl]: FINAL PREDICTION: AS
[INFO] [1775096565.014160186] [trick_logic_node_rl]: =====
```

- **Asynchronous ROS 2 Framework:**

Modular node design decouples perception, logic, and physical action, ensuring the vision pipeline and LLM speech do not block physical execution.

- **Sim-to-Real Translation:**

The central Trick Logic Node acts as the bridge, seamlessly converting the PPO policy outputs into executable Stretch 3 joint trajectories.

Detector Node:
Runs YOLOv8.
Publishes card detections.

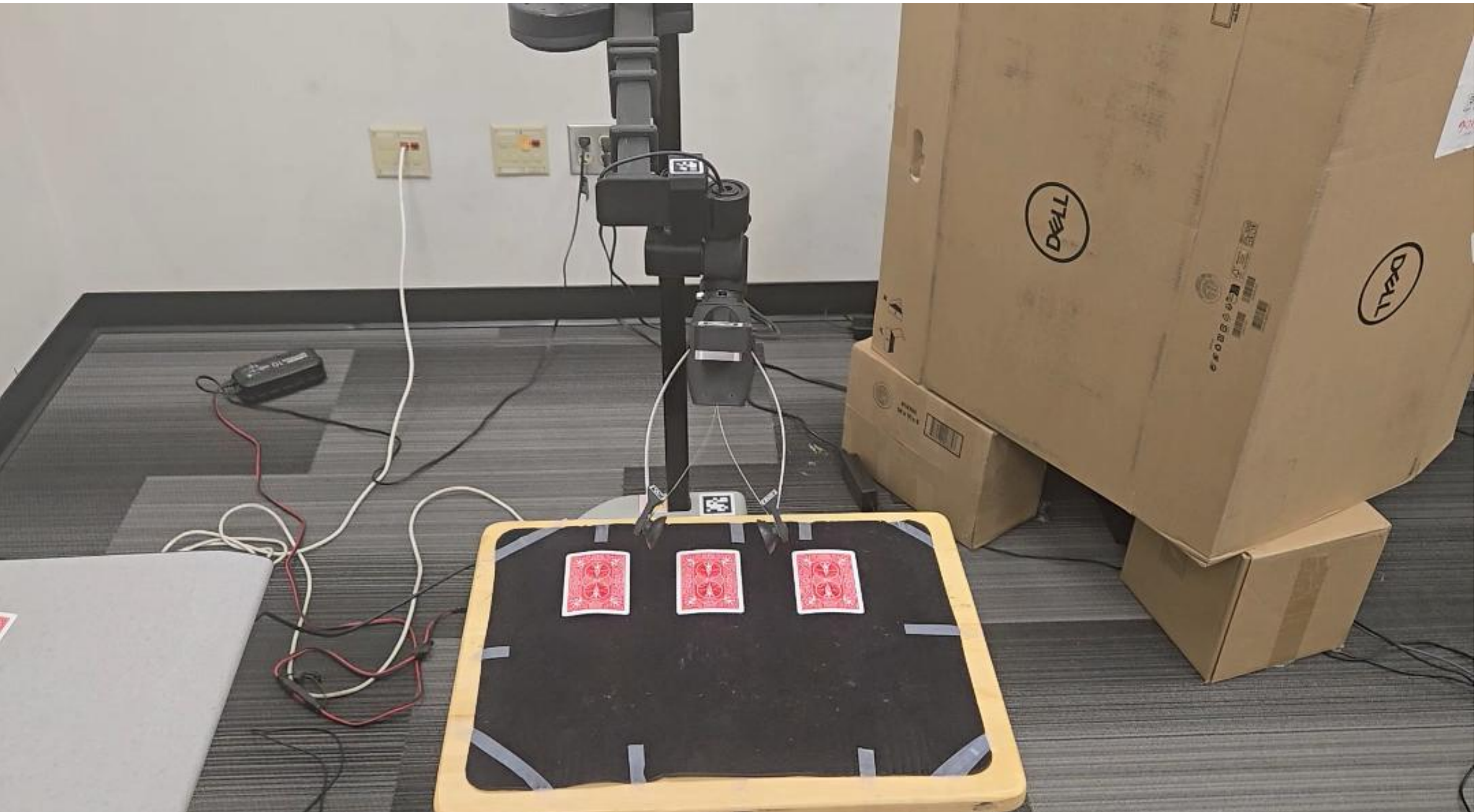


Trick Logic Node:
Loads PPO Brains 1 & 2.
Manages trick state.



Flipper Node:
Executes Stretch kinematics.
Manages LLM Speech.

Sim-to-Real Execution: Autonomous Three-Card Deception



System Evaluation (Heuristic vs. RL)

Step 1: The Heuristic Mapping (Simple 3-Card Rule: Target stays, others swap)



Initial State: [Index 0: A♠, Index 1: K♥, Index 2: Q♦]

Deduction Mapping for 3 Cards

The Heuristic (Rigid Code: If Observation -> Deduce Target)

1. **OBSERVATION:** See Q♦ at Index 1
(Implies K♥ and Q♦ swapped)

2. **OBSERVATION:** See A♠ at Index 1
(Implies A♠ and K♥ swapped)

3. **OBSERVATION:** See K♥ at Index 1
(Implies K♥ did not move)

HEURISTIC
(HARDCODED RULES)

1. **DEDUCTION:** Target is Index 0 (A♠)

2. **DEDUCTION:** Target is Index 2 (Q♦)

3. **DEDUCTION:** Target is Index 1 (K♥)

Step 2: The Breaking Point (Complex Input Causes Failure)

4. **COMPLEX OBSERVATION:**
See 4-Card Cyclic Shift
(A new, more complex movement pattern
not covered by the 3-card rule)



SYSTEM FAILURE
(Code cannot handle new data.
Requires manual reprogramming and
recompilation of the heuristic code)

Metric	Hardcoded Baseline	Two-Stage RL Framework
Human Logic Required	Yes (Explicit Programming)	No (Autonomously Discovered)
Adaptability to N-Cards	Fails (Requires Rewrite)	Succeeds (Autonomously Retrains)
Logic Inference Latency	< 1 ms	≈ 10 ms
Sim-to-Real Reliability	100%	100%

N-Card Scalability



- **The Next Milestone:**

Proving the framework generalizes beyond the 3-card baseline.

- **Physical Deployment:**

Workspace configured for a 4-card array using the Stretch 3 robot.

- **The Complexity Spike:**

Adding a single variable exponentially increases the state space, rendering 3-card heuristics obsolete.

Mechanics Step 1: The Expanded Initial State

Step 1: The Eyes (Initial State)



Index 0 Index 1 Index 2 Index 3

Observer registers starting
positions: [A♠, K♥, Q♦, J♣].

- **Establishing Ground Truth:**

Before any physical action occurs, the logic node must register the board's starting permutation via terminal initialization.

- **The State Vector:**

The state space expands. Each of the four cards is explicitly mapped to a discrete spatial index.

- **Example State:**

[Index 0: A♠,
Index 1: K♥,
Index 2: Q♦,
Index 3: J♣]

Mechanics Step 2: The Action

Step 2: The Action (Hidden Cyclic Right Shift)



User picks Q♦ (Target) at Index 2. It stays. Other cards shift cyclically: 0→1, 1→3, 3→0.

- **The New Physics:**
Transitions from a simple 2-card swap to a **Cyclic Right Shift**.
- **The Movement Rule:**
The target card stays stationary. All other cards shift right.
- **The Perception Blind Spot:**
Manipulation occurs face-down. The system knows the math of the shift, but cannot track card identities. The final board becomes a hidden state.

Mechanics Step 3: The Ambiguity Problem

Step 3: The Brain (First Flip - Ambiguous)



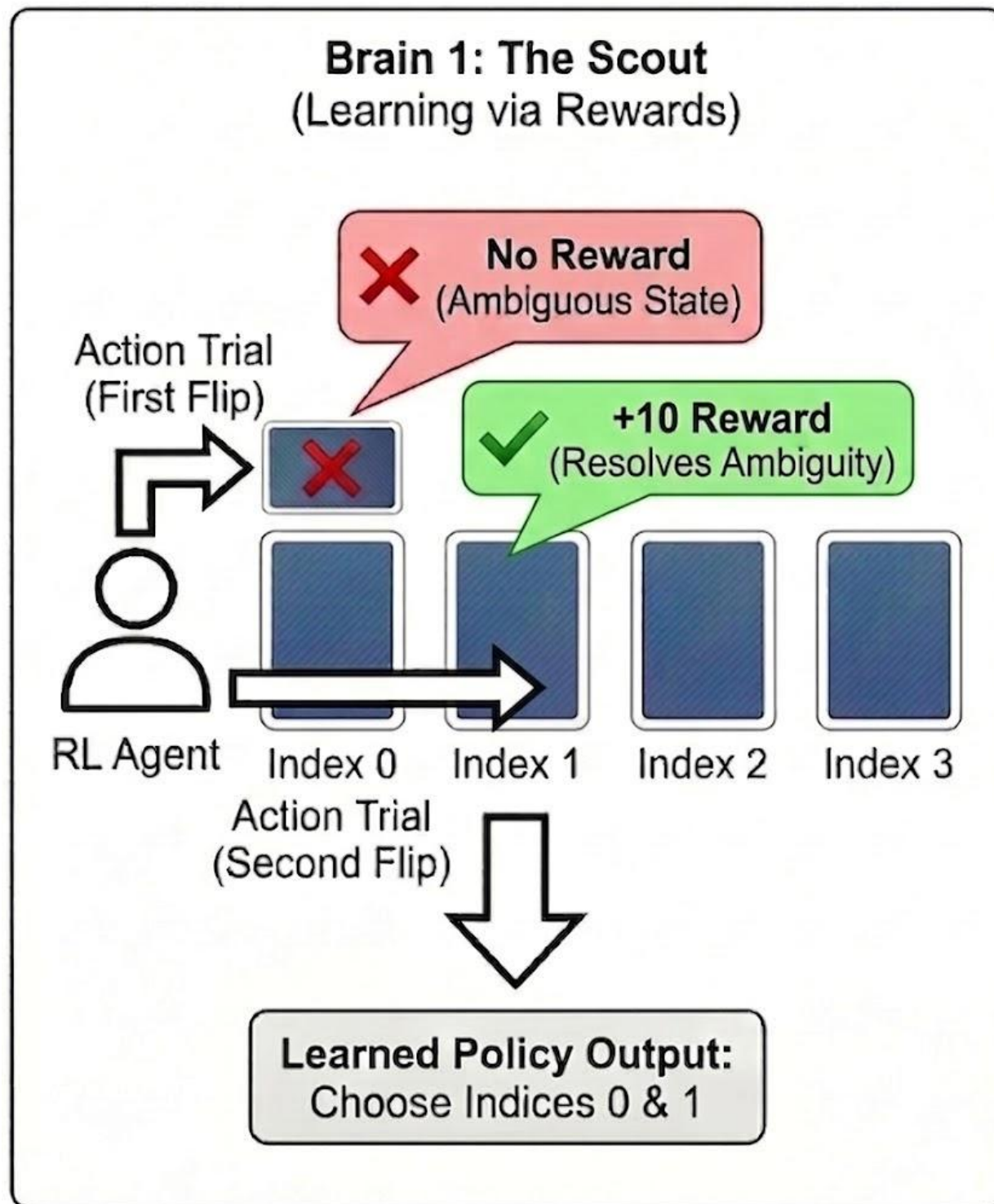
Observed J♣ at Index 0
-> Implies a cyclic shift,
but target position is
still ambiguous.

Observed A♠ at Index 1
-> Combined with J♣
at Index 0, confirms Q♦
must be at Index 2

Observer flips Index 1, sees A♠.
Two flips resolve ambiguity,
Q♦ is the Target at Index 2.

- **The Ambiguity:**
Flipping a single card yields overlapping probabilities. One observation cannot mathematically isolate the target.
- **The Multi-Step Fix:**
The system must execute a sequential, two-flip strategy to gather more data.
- **The Deduction:**
Cross-referencing both observations against the known initial state breaks the ambiguity and reveals the hidden card.

Upgrading the Scout (Multi-Step Policy)



- **Action Space Expansion:**

Upgraded from single-card flips to mapping all six possible index pairs.

- **Reward Restructuring:**

The agent receives +10 only if the chosen pair yields a completely unique observation state for all 4 possible target choices. Ambiguous pairs receive -10.

- **Autonomous Learning:**

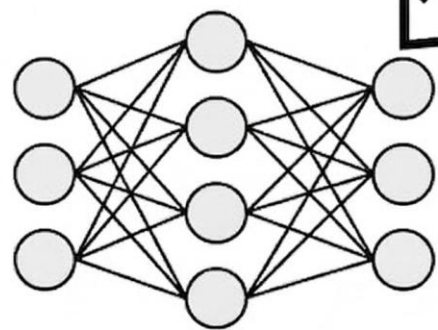
The PPO policy converges on a sequential strategy (e.g., flipping Index 0 & 1) to isolate the target mathematically.

Upgrading the Detective (Multi-Variable Inference)

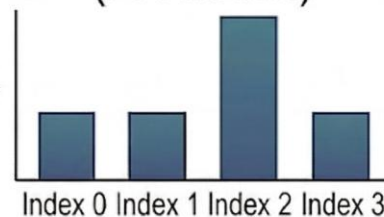
Brain 2: The Detective (Logic via Example)

KNOWN INITIAL STATE:

[Index 0: AS, Index 1: KH, Index 2: QD, Index 3: JS]



Action Probabilities (Prediction)



PPO Neural Network
(Trained on Simulated Shuffles)

FINAL PREDICTION:

The un-moved card at Index 2 (QD)
is the Target.

+1
Reward

- **Expanded Input:**

Transitions from a single visual observation to a multi-variable array (e.g., seeing both a Jack *and* an Ace).

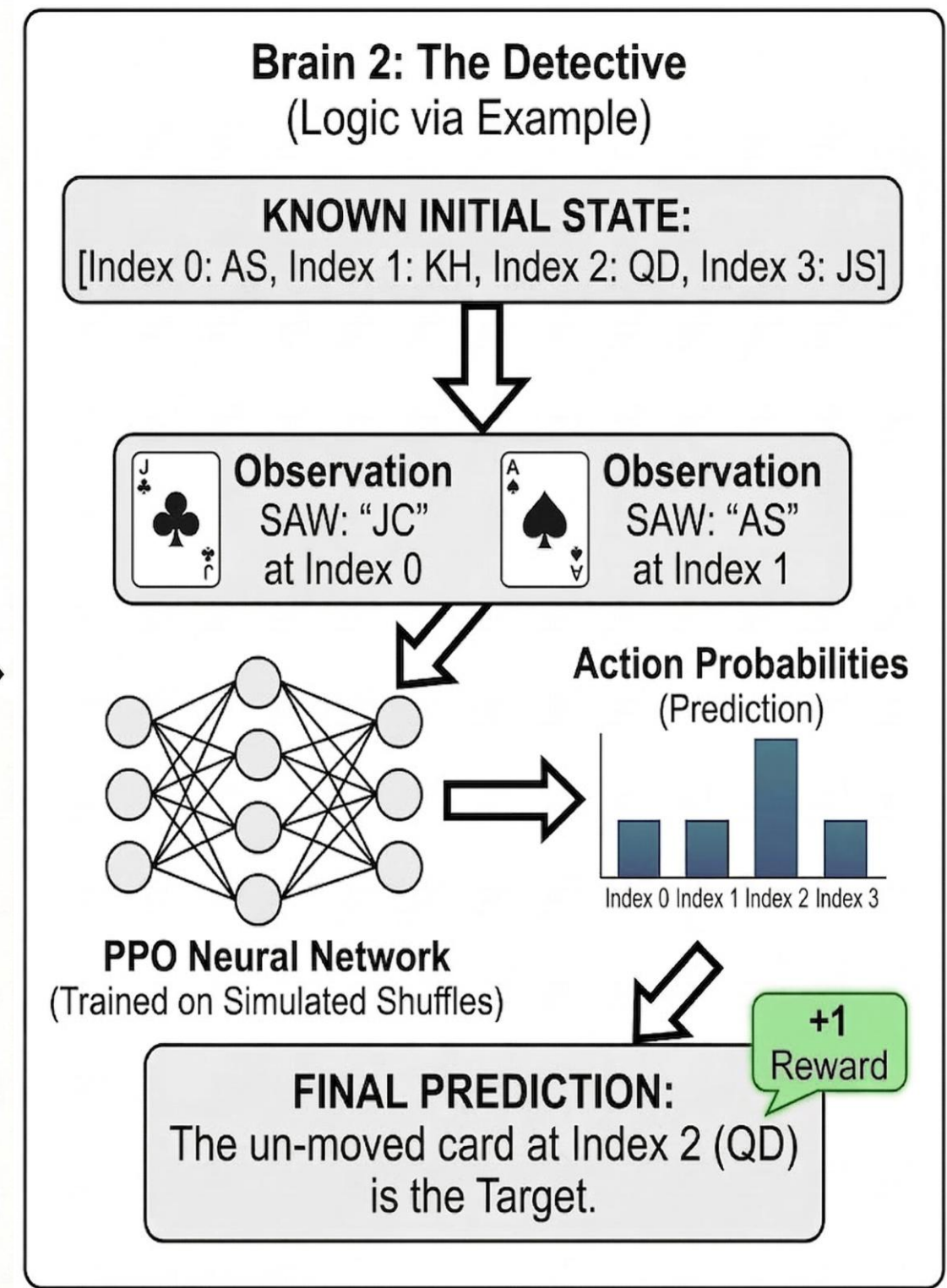
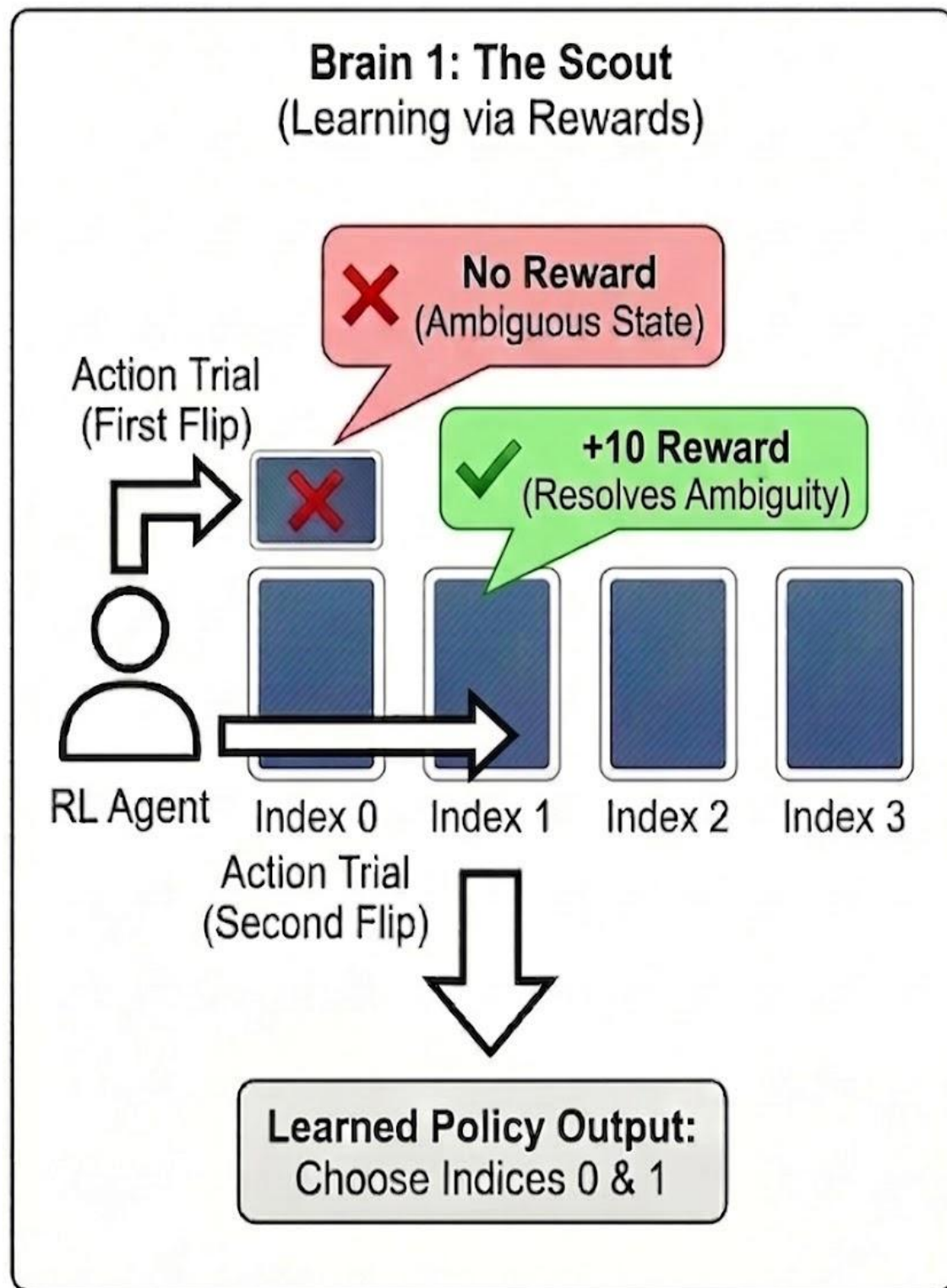
- **Simultaneous Processing:**

The neural network evaluates both physical observations concurrently against the known initial state.

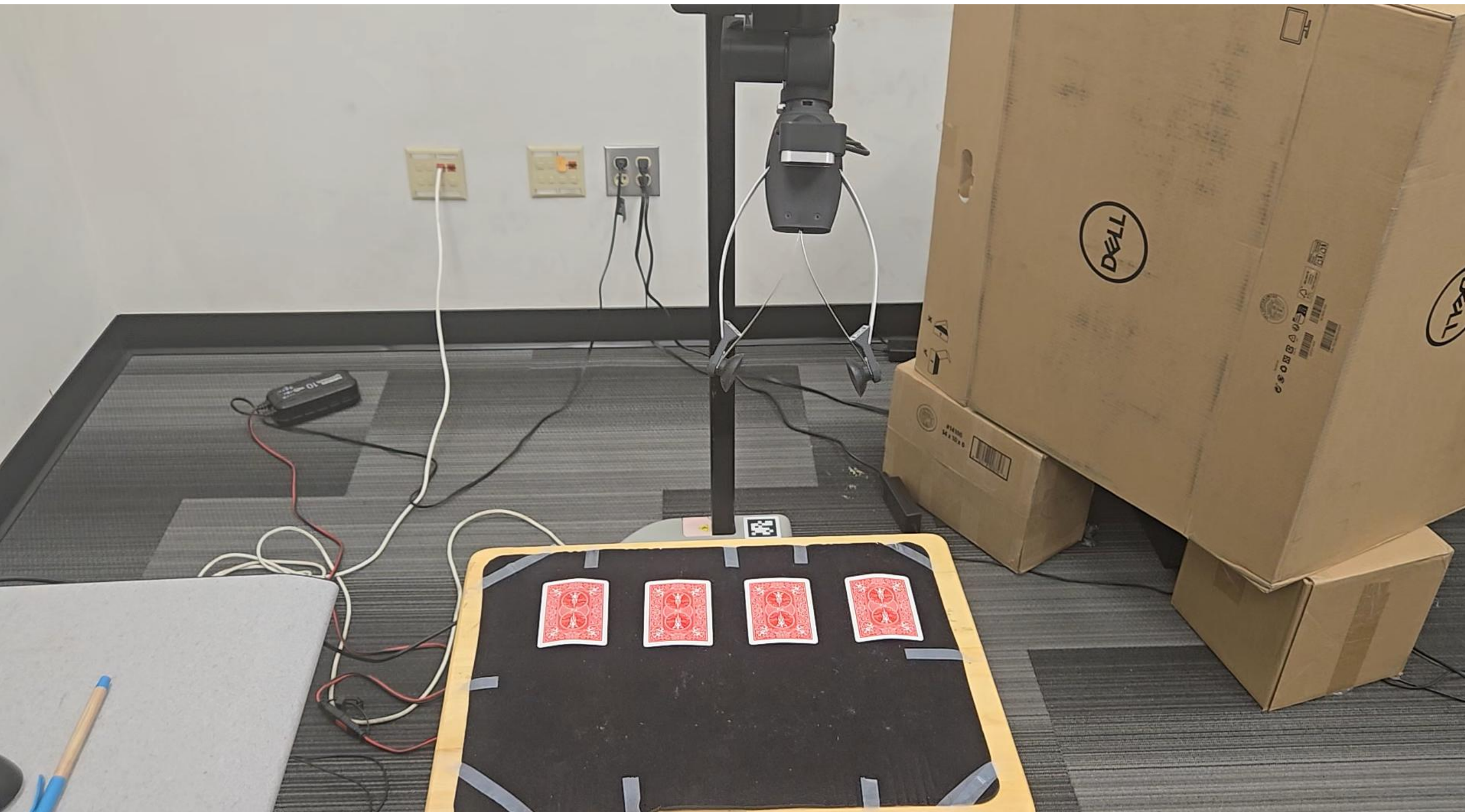
- **Target Deduction:**

Cross-referencing this combined data perfectly resolves the mathematical ambiguity, isolating the hidden card.

The Decoupled N-Card Architecture



4-Card System Demonstration



Algorithmic Validation & Performance

System Metric	3-Card Baseline	4-Card Scaled Variant
Action Space (Scout)	3 Single Indices	6 Index Pairs
State Array (Detective)	1 Observation Variable	2 Observation Variables
Inference Latency	≈ 10 ms	≈ 10 ms
Algorithmic Reliability	100% Deterministic	100% Deterministic

- **Policy Convergence:**
Models mastered the expanded state space with zero hardcoding.
- **Real-Time Efficiency:**
The decoupled inference maintains an ultra-low **~10 ms** latency.
- **100% Reliability:**
The system deterministically resolves the cyclic ambiguity every single time.

Conclusion



- **Decoupled Architecture**
- **Baseline Superiority & Scalability**
- **Sim-to-Real Reliability**
- **Interactive Autonomy**

Future Work



- **Stochastic Environments**
- **Continuous State-Action Spaces**
- **Adversarial Human Interaction**

Thank You